PORTFOLIO · 2026

Nick Lian

I take ideas to shipped software with AI agents — and set the rules that keep them from breaking.

UBC Sauder, Class of 2026 · Finance + Business Analytics

Open to anywhere in Canada

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SIX MONTHS

Four kinds of things I built

The numbers are output, not users, as of Oct 2026.

45	1,692	26	27
projects	logged updates	apps on one private database	public repos

Delivered to others

- Letterkeep
  Reference-letter system · in use by a professor
- Query Mirror
  SQL teaching explorer · no-code tool for a professor

Public

- Scholar Outflow Lab
  13 countries, recomputed weekly
- CallDelta
  Earnings-call sentiment · tone → residual signal
- Vacancy Signal
  Ghost job postings as an equity signal · 310K postings

Used every day

- 26-app personal operating system

Runs on its own

- Multi-agent content pipeline
  topic to release pack
- 10 scheduled jobs
- Guardrails and sentinels
  machine blocks dangerous operations

FOUR WORKS

What I built, and the decision inside each

WORK 1 · DELIVERED TO A PROFESSOR

Letterkeep

Reference-letter system

Roster in, students screened by grade and attendance, documents uploaded, emails out. Shipped as a web app in a day; rebuilt as a desktop app after reading the privacy law, so student data stays on the professor's machine.

3 months · 56 iterations · Python / Flask / SQLite / OCR

THE POINT **Nobody asked. I read the law and changed the architecture.**

WORK 2 · PUBLIC

nickkk-skills

14 open-source agent skills for Claude Code

Seven make a deliverable: a one-page data report, a three-statement financial model, a slide deck. Seven check or guard an agent's work: an evidence audit, a regression baseline, a git guardrail hook, and drafts with every claim traced to a file.

MIT · 13 of 14 tested in Claude Code and in Codex · NickkkLian/nickkk-skills

THE POINT **"Done, tested" is a claim until someone can check it.**

WORK 3 · RUNS ON ITS OWN

Multi-agent content pipeline

Topic to release pack across several content channels. Agents exchange files, never chat; resumable state machine, bounded retries, a QC loop that sends failures back, and a human always publishes.

32 modules · 20 skills · usage ledger reconciled to the platform bill

THE POINT **My own ledger covered 1.2% of the real bill. The bill became the source of truth.**

WORK 4 · KNOWLEDGE

Structured knowledge bases built with AI

Thousands of entries across several private libraries. AI does the entry work; every entry carries a permanent ID and a source; reuse is by pointer, never by copy.

Verbatim first · no lossy simplification · checks before and after every change

THE POINT **The hard part is the acceptance discipline, not the AI.**

Six of ten public projects

Each one is a complete, documented tool. Also public: Clipbind, Headersort, Bill Bench, and the 14 agent skills in nickkk-skills.

Query Mirror

No-code SQL teaching tool for a professor: click tables, filters and groups, the SQL appears live. Deterministic by design, no LLM.

Vanilla JS · sql.js (SQLite in WebAssembly) vendored

NickkkLian/SQL-Viz-for-Edu

Scholar Outflow Lab

Where researchers from 13 countries end up, from OpenAlex open data. Ranked by Wilson lower bound so small samples sink. Recomputed weekly.

Python stdlib scraper · GitHub Pages · launchd

NickkkLian/Scholar-Outflow-Lab

CallDelta

Management framing vs analyst pushback, by topic, across quarters. Does tone explain the residual move net of the EPS surprise?

Python · JSON schema + Pydantic · static dashboard

NickkkLian/Earnings-Call-Sentiment-Analyser

Vacancy Signal

Ghost postings in 310K listings, scored per firm. Backtest 2019–2026: the long/short lost 22.1%; both tails long roughly matched SPY. Team of three.

XGBoost · GMM · SHAP · Revelio data

NickkkLian/Ghost-Job-Detection-And-Trading-Signal

Letterkeep

Students check whether they qualify for a reference letter before they ask. The roster, grades and files stay on the professor’s own machine.

Flask · SQLite · guided tour

NickkkLian/Prof-Reference-System

ApplyLedger

A multi-region job-search tracker: prompt generators, batch resume tailoring, and the GitHub Contents API as its backend.

React 18 compiled ahead of time · GitHub Contents API

NickkkLian/Job-Tracker

HOW I WORK

Ship first, then let machines keep it honest

Most guardrail rules come from a real incident. I don't hand-write code: I define the problem, set the constraints, verify the result, and own the result.

Keeping systems reliable

- A pre-tool hook blocks dangerous operations
  force push, mass deletion, pushing while behind · 421 lines · ask by default
- Sentinels raise alarms
  database shrinks, copies drift, sessions collide
- The reviewer is never the doer
  an audit agent judges evidence only
- Three backup tiers, restores actually rehearsed

What I can do

- Take an idea to production
- Set the rules that keep systems reliable
- Python backends, web frontends, agent orchestration
  AI sessions in parallel, one repo each
- Product judgment: what is worth building, what to cut
- Business grounding: valuation, financials, backtesting
  bank valuation work · CFA Level I